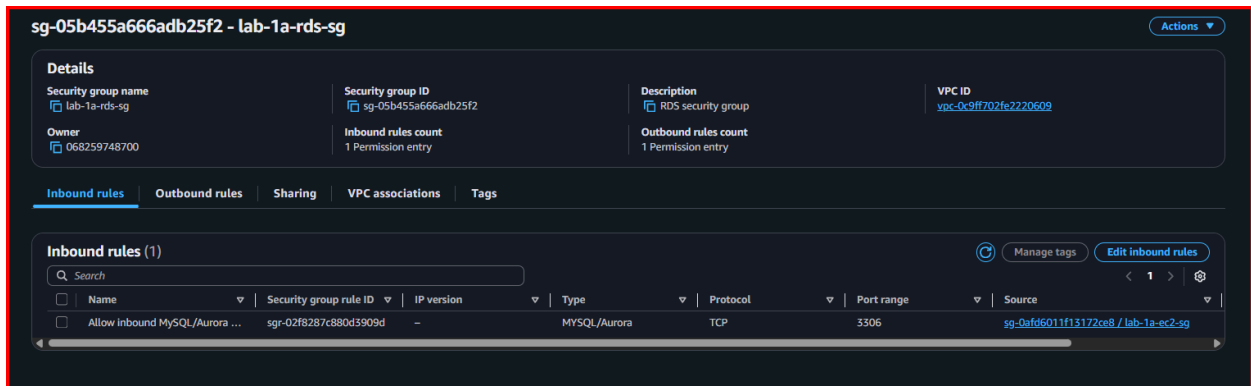


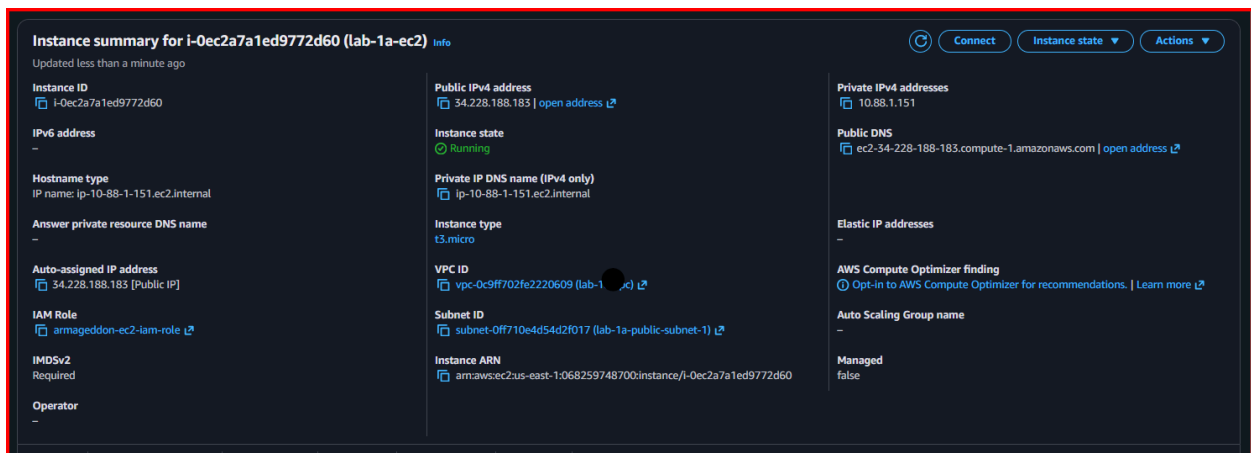
Student Deliverables:

1) Screenshot of:

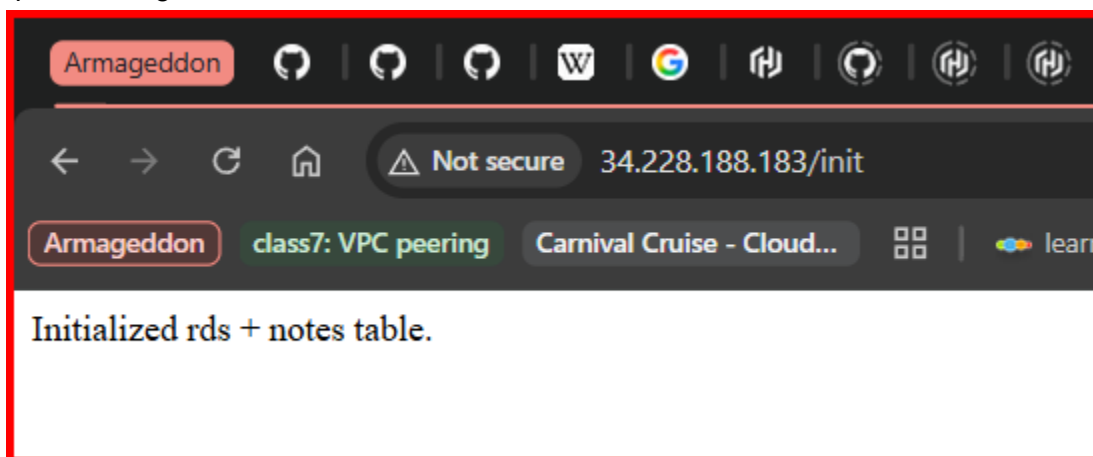
RDS SG inbound rule using source = sg-ec2-lab

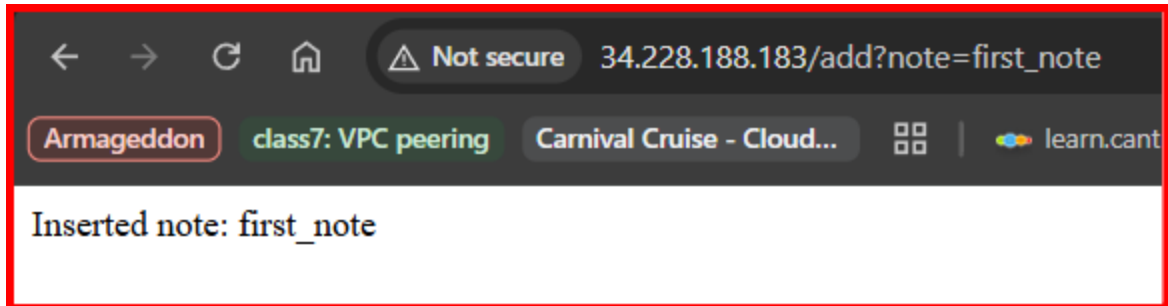


EC2 role attached

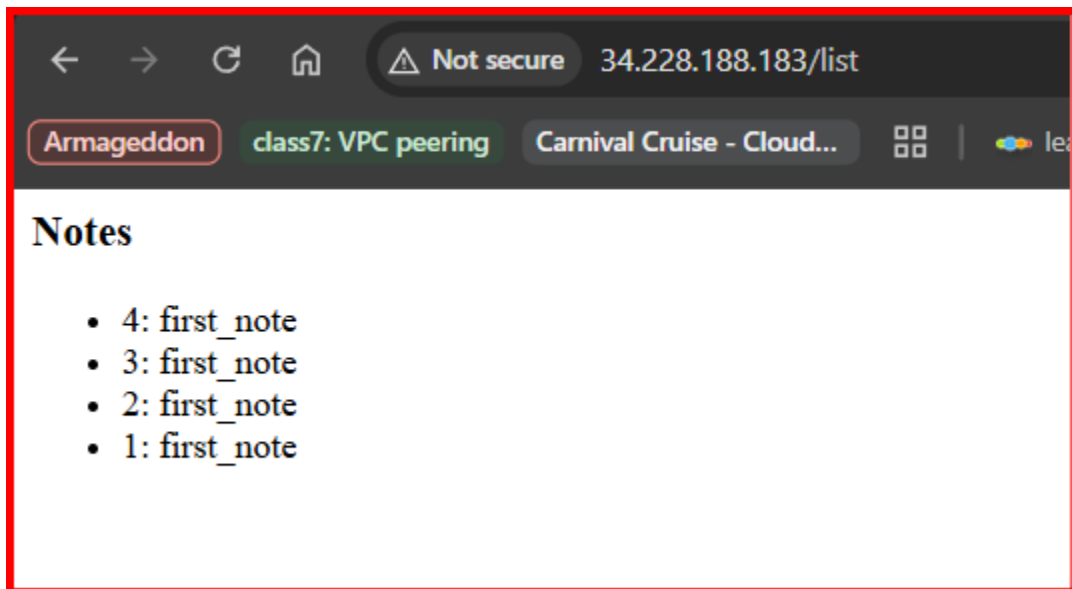


/list output showing at least 3 notes:





2.



3.

2) Short answers:

A) Why is DB inbound source restricted to the EC2 security group?

- Principle of least privilege and security best practices. Must ensure that the correct principles have access to the database.

B) What port does MySQL use?

- Port 3306

C) Why is Secrets Manager better than storing creds in code/user-data?

- To avoid your secrets from being exposed. This is especially the case if you are pushing your code into a VCS like GitHub. You do not want your secrets uploaded into the cloud.

3) Evidence for Audits / Labs (Recommended Output)

aws ec2 describe-security-groups --group-ids sg-0afd6011f13172ce8 > sg.json

```
naufv@lonbrajdife MINGW64 ~/Documents/TheowAF/Class7/AWS/Armageddon/Class-7-Armageddon/Lab-1 (mike-armageddon-branch)
aws ec2 describe-security-groups \
--group-ids sg-0afd6011f13172ce8 \
--region us-east-1 \
--output json

{
  "SecurityGroups": [
    {
      "GroupId": "sg-0afd6011f13172ce8",
      "IpPermissionsEgress": [
        {
          "IpProtocol": "tcp",
          "FromPort": 80,
          "ToPort": 80,
          "UserIdGroupPairs": [
            {
              "UserId": "068259748700",
              "GroupId": "sg-05b455a666adb25f2"
            }
          ],
          "IpRanges": [],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        },
        {
          "IpProtocol": "-1",
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ],
      "Tags": [
        {
          "Key": "Name",
          "Value": "lab-1a-ec2-sg"
        }
      ],
      "VpcId": "vpc-0c9ff702fe2220609",
      "SecurityGroupArn": "arn:aws:ec2:us-east-1:068259748700:security-group/sg-0afd6011f13172ce8",
      "OwnerId": "068259748700",
      "GroupName": "lab-1a-ec2-sg",
      "Description": "EC2 Web App security group",
      "IpPermissions": [
        {
          "IpProtocol": "tcp",
          "FromPort": 80,
          "ToPort": 80,
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        },
        {
          "IpProtocol": "tcp",
          "FromPort": 22,
          "ToPort": 22,
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        },
        {
          "IpProtocol": "tcp",
          "FromPort": 3389,
          "ToPort": 3389,
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ]
    }
  ]
}
```

aws rds describe-db-instances --db-instance-identifier lab-1ards > rds.json

```
MINGW64:/c:/Users/mmauv/Documents/TheoWAF/class7/AWS/Armageddon/Class-7-Armageddon/lab-1 (mike-armageddon-branch)
$ aws rds describe-db-instances \
--db-instance-identifier lab-1ards \
--region us-east-1 \
--output json
{
  "DBInstances": [
    {
      "DBInstanceIdentifier": "lab-1ards",
      "DBInstanceClass": "db.t3.micro",
      "Engine": "mysql",
      "DBInstanceStatus": "available",
      "MasterUsername": "admin",
      "DBName": "rds",
      "Endpoint": {
        "Address": "lab-1ards.cwuz3uwlwom.us-east-1.rds.amazonaws.com",
        "Port": 3306,
        "HostedZoneId": "Z2R221UGPM61AM"
      },
      "AllocatedStorage": 20,
      "InstanceCreateTime": "2026-01-17T08:23:09.141000+00:00",
      "PreferredBackupWindow": "10:22-10:52",
      "BackupRetentionPeriod": 0,
      "DBSecurityGroups": [],
      "VpcSecurityGroups": [
        {
          "VpcSecurityGroupId": "sg-05b455a866adb25f2",
          "Status": "active"
        }
      ],
      "DBParameterGroups": [
        {
          "DBParameterGroupName": "default:mysql8.0",
          "ParameterApplyStatus": "in-sync"
        }
      ],
      "AvailabilityZone": "us-east-1a",
      "DBSubnetGroup": {
        "DBSubnetGroupName": "armageddon-rds-subnet-group",
        "DBSubnetGroupDescription": "Managed by Terraform",
        "VpcId": "vpc-0c9ff702fe2220609",
        "SubnetGroupStatus": "Complete",
        "Subnets": [
          {
            "SubnetIdentifier": "subnet-06adea9d581f0a881",
            "SubnetAvailabilityZone": {
              "Name": "us-east-1c"
            },
            "SubnetOutpost": {},
            "SubnetStatus": "Active"
          },
          {
            "SubnetIdentifier": "subnet-057b6c8505f140abf",
            "SubnetAvailabilityZone": {
              "Name": "us-east-1b"
            },
            "SubnetOutpost": {},
            "SubnetStatus": "Active"
          },
          {
            "SubnetIdentifier": "subnet-08514a0a4ab347a08",
            "SubnetAvailabilityZone": {
              "Name": "us-east-1a"
            },
            "SubnetOutpost": {},
            "SubnetStatus": "Active"
          }
        ]
      },
      "PreferredMaintenanceWindow": "tue:07:26-tue:07:56",
      "UpgradeRolloutOrder": "second",
      "PendingModifiedValues": {},
      "MultiAZ": true,
      "EngineVersion": "8.0.43",
      "AutoMinorVersionUpgrade": true,
      "ReadReplicaDBInstanceIdentifiers": [],
      "LicenseModel": "general-public-license",
      "StorageThroughput": 0,
      "OptionGroupMemberships": [
        {
          "OptionGroupName": "default:mysql8-0",
          "Status": "in-sync"
        }
      ],
      "SecondaryAvailabilityZone": "us-east-1b",
      "PubliclyAccessible": false,
      "StorageType": "gp2",
      "DBInstancePort": 0,
      "StorageEncrypted": false,
      "DBResourceIdentifier": "db-1H9PWA346TLKXELP36YFTECY",
      "CACertificateIdentifier": "rds-ca-rsa2048-g1",
      "DomainMemberships": [],
      "CopyTagsToSnapshot": false,
      "MonitoringInterval": 0,
      "DBInstanceArn": "arn:aws:rds:us-east-1:068259748700:db:lab-1ards",
      "IAMDatabaseAuthenticationEnabled": false,
      "DatabaseInsightsMode": "standard",
      "PerformanceInsightsEnabled": false,
      "DeletionProtection": false,

```

```

    },
    "CustomerOwnedIpEnabled": false,
    "NetworkType": "IPv4",
    "ActivityStreamStatus": "stopped",
    "BackupTarget": "region",
    "CertificateDetails": {
      "CAIdentifier": "rds-ca-rsa2048-g1",
      "ValidTill": "2027-01-17T08:21:50+00:00"
    },
    "DedicatedLogVolume": false,
    "IsStorageConfigUpgradeAvailable": false,
    "EngineLifecycleSupport": "open-source-rds-extended-support"
  }
}
```

aws secretsmanager describe-secret --secret-id lab-1a/rds/mysql01 > secret.json

```
mmauv@lonbrajdife MINGW64 ~/Documents/TheoWAF/class7/AWS/Armageddon/Class-7-Armageddon/lab-1 (mike-armageddon-branch)
$ aws secretsmanager describe-secret \
  --secret-id lab-1a/rds/mysql01 \
  --region us-east-1 \
  --output json
{
  "ARN": "arn:aws:secretsmanager:us-east-1:068259748700:secret:lab-1a/rds/mysql01-1KZsmP",
  "Name": "lab-1a/rds/mysql01",
  "LastChangedDate": "2026-01-17T03:32:57.630000-05:00",
  "LastAccessedDate": "2026-01-16T19:00:00-05:00",
  "VersionIdsToStages": {
    "terraform-20260117083257574400000006": [
      "AWSCURRENT"
    ]
  },
  "CreateDate": "2026-01-17T03:19:25.600000-05:00"
}
```

aws ec2 describe-instances --instance-ids i-0ec2a7a1ed9772d60 > instance.json

```
mmauv@lonbrajdife MINGW64 ~/Documents/TheoWAF/class7/AWS/Armageddon/Class-7-Armageddon/lab-1 (mike-armageddon-branch)
$ aws ec2 describe-instances \
  --instance-ids i-0ec2a7a1ed9772d60 \
  --region us-east-1 \
  --query "Reservations[].Instances[].IamInstanceProfile.Arn" \
  --output text
arn:aws:iam::068259748700:instance-profile/armageddon-instance-profile
```

aws iam list-attached-role-policies --role-name armageddon-ec2-iam-role > role-policies.json

```
mmauv@lonbrajdife MINGW64 ~/Documents/TheoWAF/class7/AWS/Armageddon/Class-7-Armageddon/lab-1 (mike-armageddon-branch)
$ aws iam list-attached-role-policies \
  --role-name armageddon-ec2-iam-role \
  --output json
{
  "AttachedPolicies": [
    {
      "PolicyName": "armageddon-iam-policy",
      "PolicyArn": "arn:aws:iam::068259748700:policy/armageddon-iam-policy"
    }
  ]
}
```

Then Answer:

Why each rule exists:

- http → ec2-sg: provides access to the notes app.
- ssh → ec2-sg: (optional) to access via cli
- ec2-sg → rds-sg: inbound rule for the rds-sg is restricted to port 3306 which only allows the ec2-sg to have access. This allows the flask app to insert the notes into the database

What would break if removed:

- We would not have a definitive means to access the notes app. In the worst case scenario, anyone could access the database

Why broader access is forbidden:

- Principle of least privilege. To allow only the necessary principle (ec2) to have access to the necessary it must have access to. This includes the IAM permissions attached to the ec2.
- To protect the secrets stored in secrets manager

Why this role exists:

- IAM provides ec2 the necessary privileges to read the secrets in secrets manager. You do not want the ec2 to do anything beyond what you intend for it to do.

Why it can read this secret:

- Because the IAM policy allows the ec2 to read the secret

Why it cannot read others:

- Because the IAM policy does not allow it to do so.